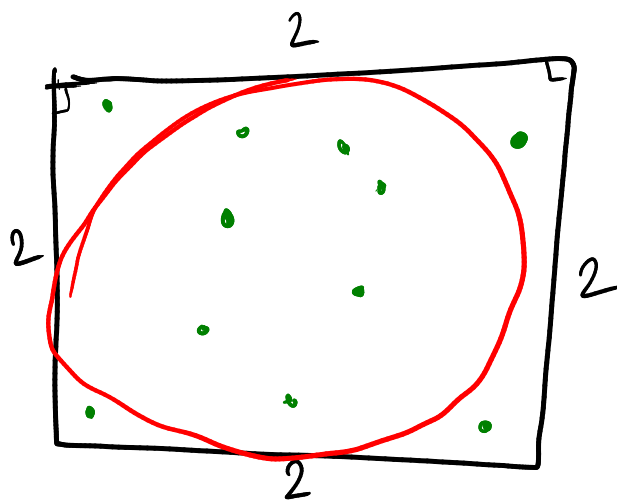




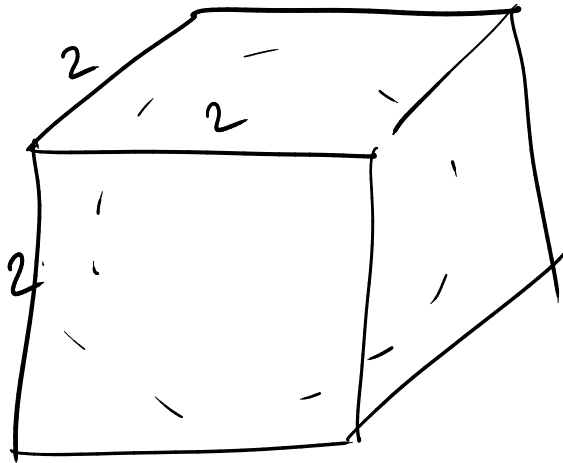
$$P(\text{inside}) \approx \frac{\# \text{ inside}}{\# \text{ total}}$$

$$\frac{\text{Area inside}}{\text{Area total}} = \frac{\# \text{ inside}}{\# \text{ total}}$$

$$P = \frac{\pi \times 1^2}{2 \times 2} = \frac{\text{in}}{\text{to}}$$

$$\pi \Rightarrow \boxed{4 \times \frac{\text{in}}{\text{to}}}$$

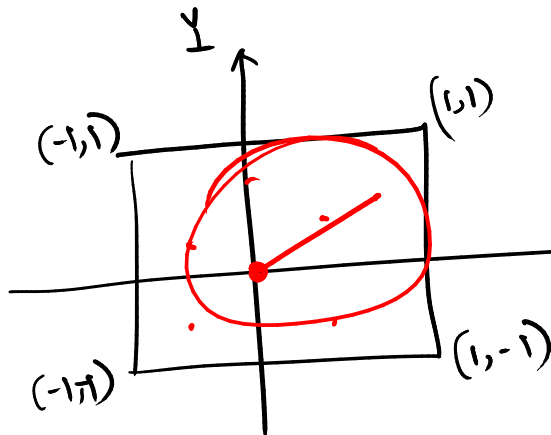
$$\pi = 4P$$



$$P = \frac{\frac{4}{3}\pi \times 1^3}{2^3} = \frac{\text{in}}{\text{to}}$$

$$P = \frac{\pi}{6} = \frac{\text{in}}{\text{to}}$$

$$\pi = 6P = 6 \times \frac{\text{in}}{\text{to}}$$



$$N \rightarrow [0, 1]$$

$$x = 2U - 1$$

$$y = 2V - 1$$

$$d = \sqrt{x^2 + y^2}$$

$$P = 1$$

③ RV with parameter $p \rightarrow X$

$$X = \begin{cases} 1 & ; p \\ 0 & ; 1-p \end{cases}$$

$$\boxed{\begin{array}{c} \text{RV} \\ \square + \square = 5 \\ \downarrow \\ \text{BRV} \dots \end{array}}$$

$$E[X], E[X^2], \text{Var}(X), \text{SD}(X) \rightarrow \phi(t) = E[e^{tx}] \rightarrow$$

$$\mu = E[X] = 1 \cdot p + 0 \cdot (1-p) = p$$

$$E[X^2] = 1^2 \cdot p + 0^2 \cdot (1-p) = p$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$$

$$\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)} = \sqrt{p(1-p)}$$

$$\begin{aligned} \text{Var}(cX) &= E[c^2 X^2] - (E[cX])^2 \\ &= c^2 E[X^2] - c^2 (E[X])^2 \\ &= c^2 \text{Var}(X) \end{aligned}$$

$$\text{SD}(cX) = c \text{SD}(X)$$

$$\begin{aligned} \text{Var}(X_1 + X_2) &= E[(X_1 + X_2)^2] - (E[X_1 + X_2])^2 \\ &= E[X_1^2] + E[X_2^2] + 2E[X_1 X_2] - (E[X_1])^2 - (E[X_2])^2 - 2E[X_1]E[X_2] \\ &= \text{Var}(X_1) + \text{Var}(X_2) \end{aligned}$$

$$\text{Var}\left(\frac{1}{N}(X_1 + X_2 + X_3 + \dots + X_N)\right) = \frac{1}{N^2} (\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_N))$$

$$\text{Var}\left(\frac{X_1 + X_2 + X_3 + \dots + X_N}{N}\right) = \frac{\text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) + \dots + \text{Var}(X_N)}{N^2}$$

$$SD\left(\frac{x_1 + x_2 + \dots + x_N}{N}\right) = \sqrt{\frac{\text{Var}(x_1) + \text{Var}(x_2) + \dots + \text{Var}(x_N)}{N^2}}$$

$$= \frac{\sqrt{SD(x_1)^2 + SD(x_2)^2 + \dots + SD(x_N)^2}}{N}$$

gk $\rightarrow x_1, x_2, \dots, x_N \rightarrow$ BRV with param p

$$SD\left(\frac{x_1 + x_2 + \dots + x_N}{N}\right) = \frac{\sqrt{p(1-p) + p(1-p) + \dots + p(1-p)}}{N}$$

$$= \frac{\sqrt{N p(1-p)}}{N} = \sqrt{\frac{p(1-p)}{N}}$$

$$E\left[\frac{x_1 + x_2 + \dots + x_N}{N}\right] = \frac{E[x_1] + E[x_2] + \dots + E[x_N]}{N}$$

$$= \frac{p + p + \dots + p}{N}$$

$$= \frac{Np}{N} = p$$

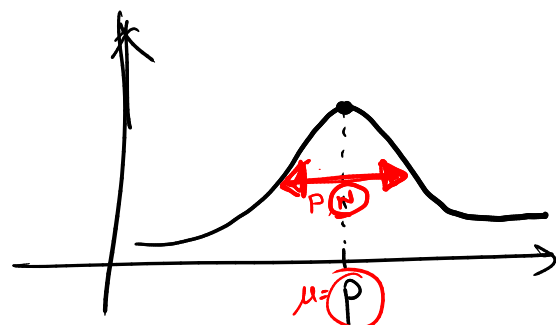
$Y = \frac{x_1 + x_2 + \dots + x_N}{N}$

Red arrows point from x_1 and x_2 to $BRV(p)$.

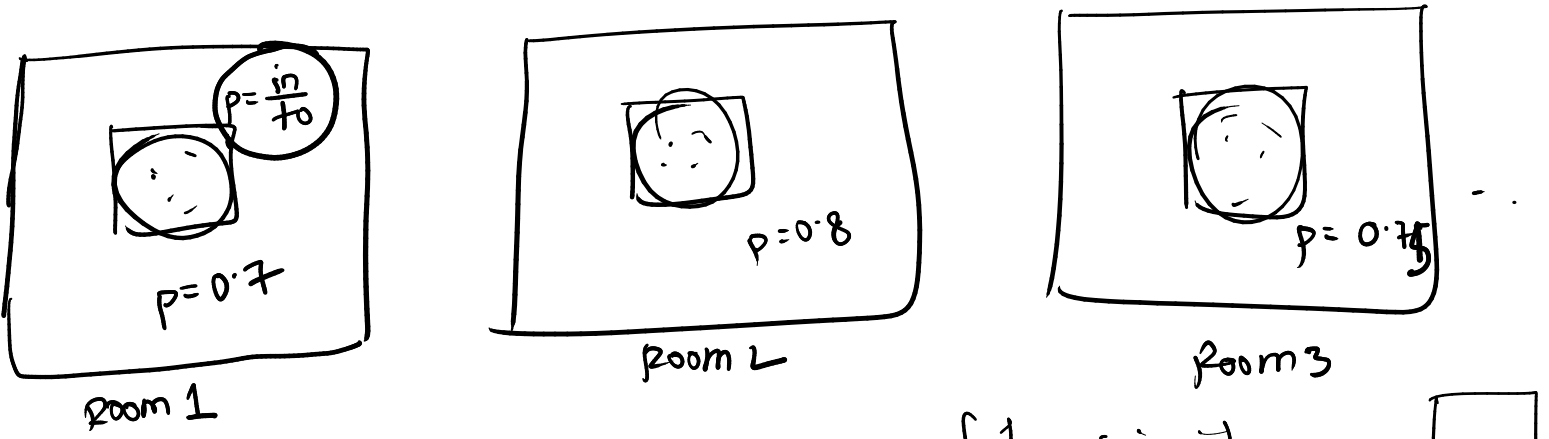
$\mu = p, \sigma = \sqrt{\frac{p(1-p)}{N}}$

Red arrows point from μ and σ to the normal distribution graph.

$\rightarrow$ normally dist μ, σ



$N = 100$

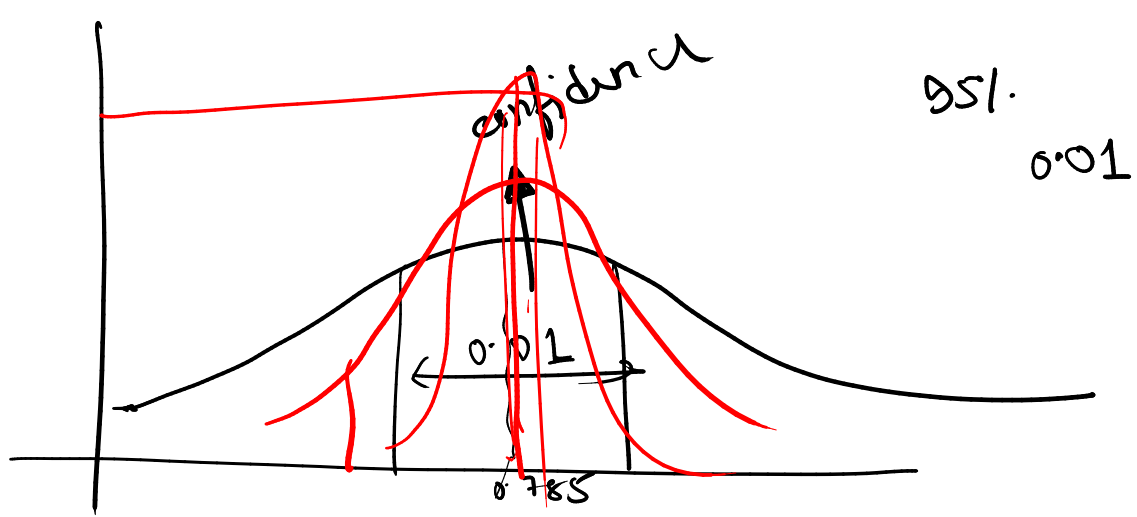
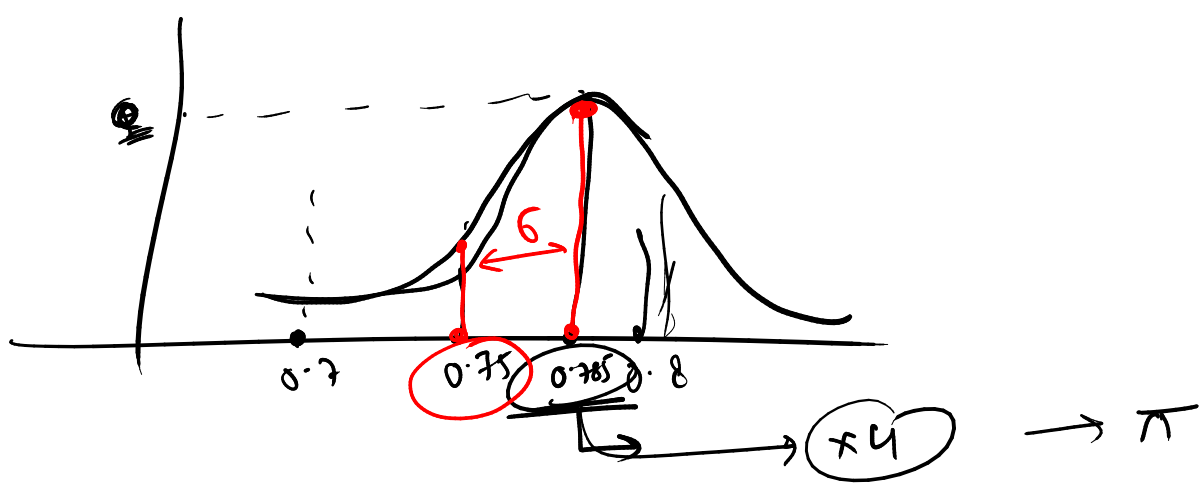


$x = 1$ if random point inside $\sigma = \begin{cases} 1 & ; \text{inside} \\ 0 & ; \text{outside} \end{cases}$

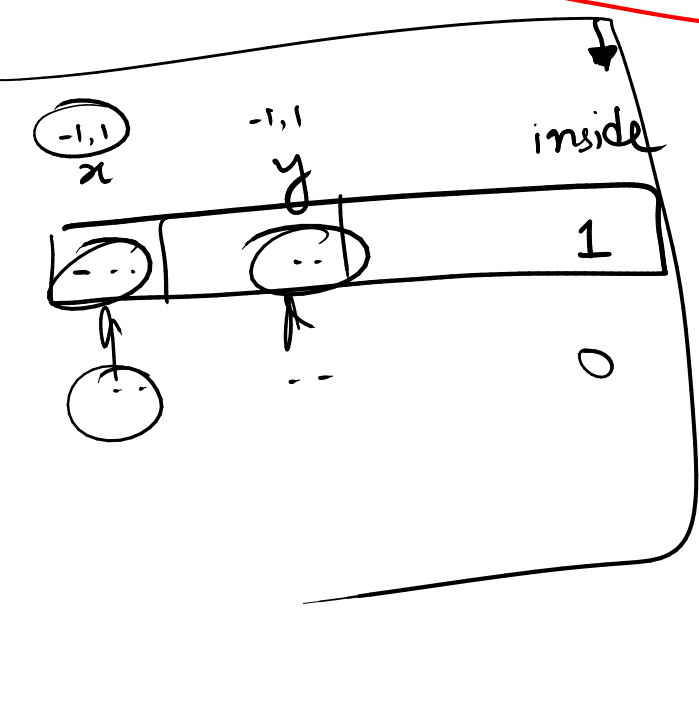
$\bar{y} = \frac{x_1 + \dots + x_N}{N}$

$N = 100$

$N \geq 30$

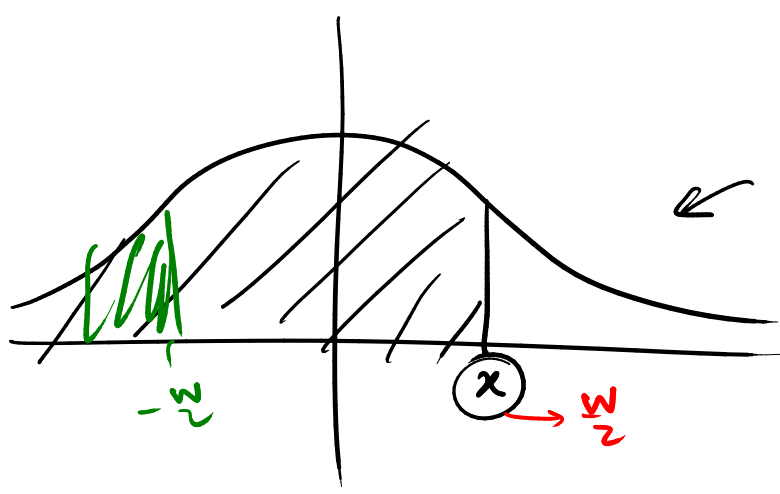


Room 1



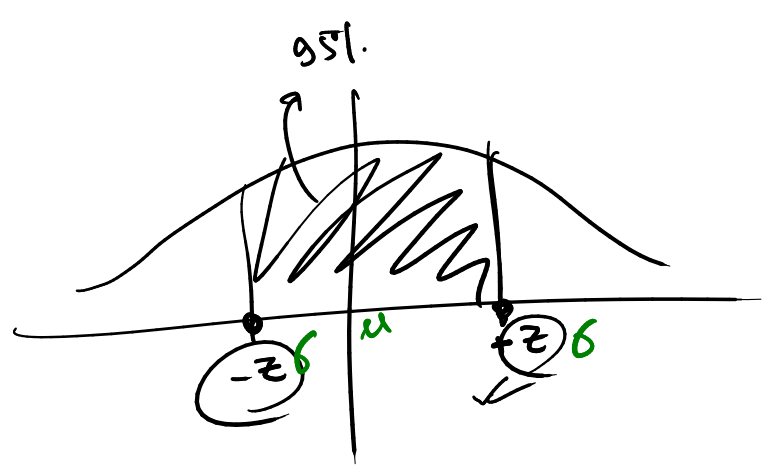
$P = \square \rightarrow \text{confidence} = \square$
 $\pi = 4 \times P$
 $w = 0.01$

$N = 100$

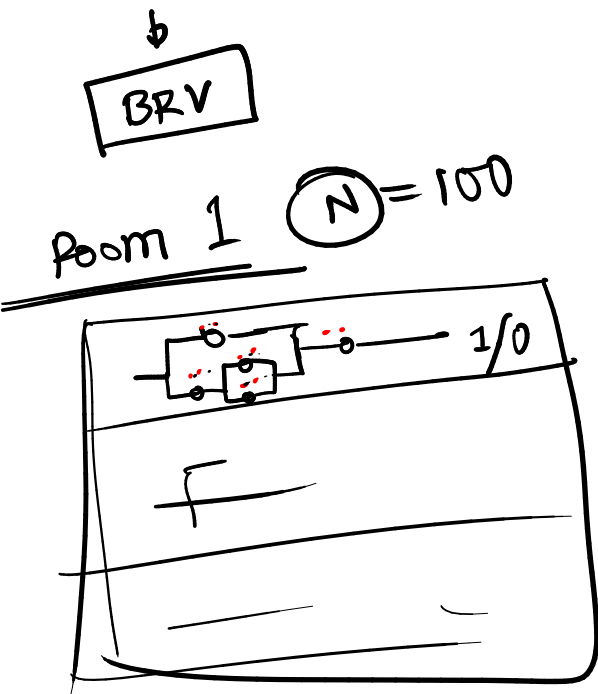
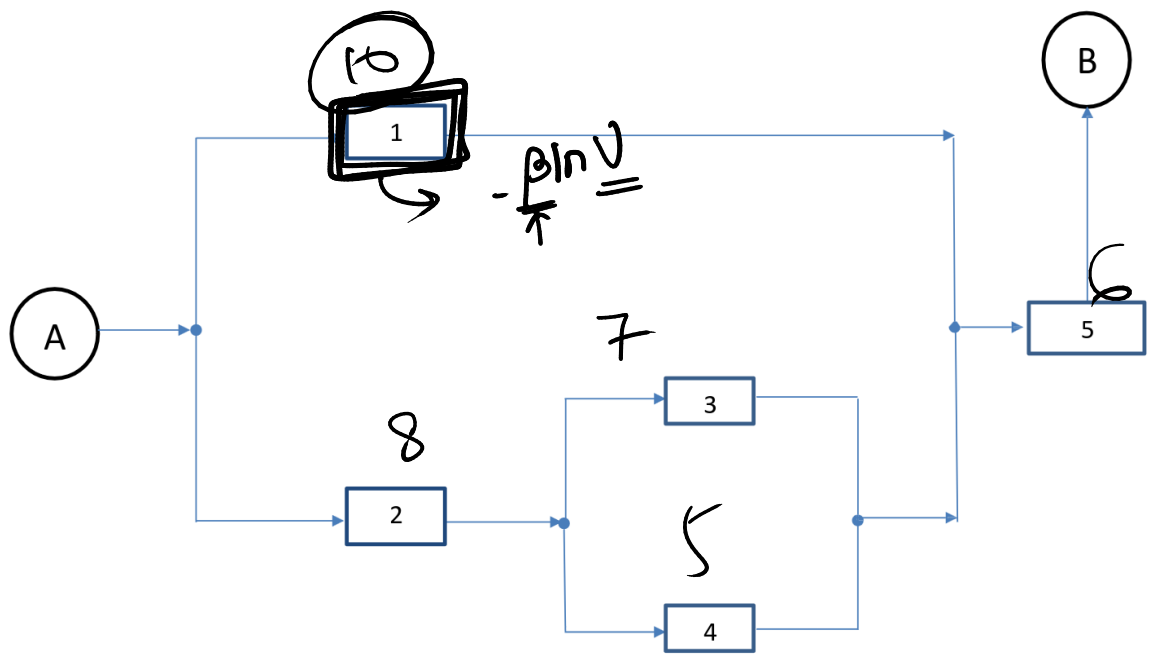


$w = 0.01, c = 95\%$

$N = ?$



$z_0 = \frac{w}{2}$
 $\frac{H}{4}$
 $\frac{P(1-P)}{N}$
 $\frac{z}{2}$
 $\square \rightarrow \square$



100

70

P

1	2	3	4	5
..	..	..	..	..
-	-	-	-	-

S	S27
...	1
	6
	1
	1

